

Practical Fixed-Income Demos

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These executable demonstrations connect the lecture models to realistic analyst and portfolio-management workflows. Inputs are clearly labeled as observed or illustrative, calculations are reproducible, and each demo includes interpretation and validation checks.

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	7	Curve construction, bootstrapping, and Hull–White calibration	From bond prices to Hull–White future yield curves
	7	Vasicek parameter interpretation and rate scenarios	Vasicek scenario workflow
	8	Callable-bond valuation, option cost, OAS, and effective risk	Callable-bond and OAS workflow
	9	DV01-based Treasury futures hedging and residual risks	Treasury futures hedging workflow
	10	Spread duration, DTS, carry, expected loss, and stress tests	Credit-spread risk workflow
	11	Treasury futures options and asymmetric portfolio protection	Interest-rate options workflow

The examples are teaching tools rather than trading recommendations. Production valuation requires current market conventions, instrument terms, liquidity inputs, and independent model validation.